

PREVENTIVE MAINTENANCE REPORT

Munich Plant 02 — Manufacturing Cell A | Components Division

Reporting period: March 2025
Issue date: 31/03/2025
Maintenance Manager: Dipl.-Ing. Thomas Bauer
Shift: Day Shift (06:00 – 14:00)
Line status: FULLY OPERATIONAL

1. Executive Summary

During March 2025, Manufacturing Cell A operated under nominal conditions throughout the entire planned production period. No unscheduled stoppages were recorded. All preventive maintenance activities were carried out according to plan PM-2025-Q1. The Overall Equipment Effectiveness (OEE) reached 95.1%, exceeding the company target of 93%. A total of 18,740 components were produced and inspected, with a first-pass yield of 99.7%.

Key Performance Indicators (KPIs)

KPI	Target	Actual	Status
OEE (%)	≥ 93%	95.1%	PASS
Availability (%)	≥ 97%	98.4%	PASS
Performance (%)	≥ 96%	96.9%	PASS
Quality / FPY (%)	≥ 99.5%	99.7%	PASS
MTBF (hours)	≥ 350	412	PASS
MTTR (hours)	≤ 2.0	1.3	PASS
Unscheduled downtime	0 events	0 events	PASS
Total parts produced	≥ 18,000	18,740	PASS

2. Maintenance Activities Performed

The following table lists all preventive maintenance interventions executed in March 2025, organised by workstation and intervention type. All tasks were completed within the allocated maintenance windows without impact on production.

Date	Workstation	Intervention	Technician	Duration (h)	Notes
04/03/2025	WS01 – CNC Milling	Tool holder inspection + torque check	K. Hoffmann	1.0	All holders within spec
06/03/2025	WS02 – Lathe	Chuck jaw replacement & alignment	T. Bauer	1.5	Runout: 0.003 mm
11/03/2025	WS03 – Hydraulic Press	Hydraulic fluid analysis & top-up	J. Mueller	0.5	ISO VG 46, viscosity OK
13/03/2025	WS04 – Assembly Robot	Joint backlash measurement + lube	S. Fischer	2.0	All axes within tolerance
18/03/2025	WS05 – CMM Inspection	Probe qualification & temp. check	L. Wagner	1.0	Probe error: 0.0012 mm
20/03/2025	WS01 – CNC Milling	Spindle bearing vibration analysis	K. Hoffmann	1.5	0.17 mm/s — nominal

Date	Workstation	Intervention	Technician	Duration (h)	Notes
25/03/2025	WS02 – Lathe	Belt tension + gearbox oil change	T. Bauer	1.0	Oil: Shell Tonna S3 M68
27/03/2025	WS03 – Hydraulic Press	Seal kit inspection on main cylinder	J. Mueller	1.5	No leakage detected
28/03/2025	WS04 – Assembly Robot	Full functional test + I/O check	S. Fischer	2.0	All 47 I/O points OK

3. Sensor Statistics Summary — March 2025

All sensor readings remained within operational thresholds as defined in technical specification TEC-CELL-A-003. Mean values and standard deviations are derived from approximately 11,500 data points collected over 8-hour day shifts.

Workstation	Sensor	Mean	Std	Min	Max	Unit	Status
WS01 – CNC Milling	Spindle Speed	4500	15.1	4450	4551	RPM	PASS
WS01 – CNC Milling	Feed Rate	350.1	3.9	338	362	mm/min	PASS
WS01 – CNC Milling	Coolant Temp	24.5	0.31	23.5	25.4	°C	PASS
WS01 – CNC Milling	Motor Current	18.2	0.41	17.1	19.4	A	PASS
WS02 – Lathe	Spindle Speed	2800	19.8	2740	2861	RPM	PASS
WS02 – Lathe	Cutting Force	620.3	7.9	597	643	N	PASS
WS02 – Lathe	Vibration	0.181	0.006	0.163	0.199	mm/s	PASS
WS03 – Hydraulic Press	Press Force	850.1	5.9	832	869	kN	PASS
WS03 – Hydraulic Press	Hydraulic Pres.	220.1	2.6	213	228	bar	PASS
WS03 – Hydraulic Press	Oil Temperature	45.0	0.82	42.7	47.1	°C	PASS
WS04 – Assembly Robot	Joint Torque	38.5	0.61	36.7	40.2	Nm	PASS
WS04 – Assembly Robot	Cycle Time	12.8	0.15	12.3	13.2	s	PASS
WS04 – Assembly Robot	Position Error	0.040	0.002	0.034	0.047	mm	PASS
WS05 – CMM Inspection	Ambient Temp.	20.0	0.10	19.7	20.3	°C	PASS
WS05 – CMM Inspection	Humidity	48.0	0.51	46.6	49.4	%RH	PASS

4. Upcoming Maintenance — April 2025

Planned Date	Workstation	Intervention	Priority
02/04/2025	WS01 – CNC Milling	Spindle oil change + filter replacement	High
08/04/2025	WS02 – Lathe	Turret indexing accuracy verification	Medium
14/04/2025	WS03 – Hydraulic Press	Pressure relief valve re-calibration	High
22/04/2025	WS04 – Assembly Robot	TCP calibration + teach-point check	High
29/04/2025	WS05 – CMM Inspection	Full probe system re-qualification	Medium
30/04/2025	Cell A (all stations)	Planned semi-annual shutdown (6 hours)	High

5. Sign-off

Role	Name	Signature	Date
Maintenance Manager	Dipl.-Ing. Thomas Bauer	_____	31/03/2025
Production Manager	Dr. Anna Schreiber	_____	31/03/2025
Plant Director	Dr.-Ing. Hans Richter	_____	31/03/2025